



1. Setup & Motivation

Score-based black-box attacks observe the model probabilities but not gradients. Standard untargeted attacks reduce only the true-class confidence:

$$\mathcal{L}_{\text{untarg}}(x', y) = P(y | x') \quad \text{or} \quad \log P(y | x').$$

This objective frees probability mass without deciding which non-true class should receive it. The attack may therefore drift across several class basins before a misclassification is reached.

Goal. Use the attack trajectory itself to choose a viable adversarial target, then commit early enough to avoid wasting queries on class drift.

Why this gap exists. Targeted attacks remove drift but require a target before the attack starts. Margin losses track a competitor dynamically, but many score-based attacks are implemented with probability or CE losses. OTS is a wrapper for exactly that case.

2. Method: Opportunistic Target Selection

OTS leaves the proposal mechanism unchanged. It modifies only the loss schedule after a short untargeted prefix.

Algorithm.

1. Run the original untargeted attack for T iterations.
2. Lock onto the leading non-true class

$$t = \arg \max_{k \neq y} P(k | x'_T).$$

3. Continue the same attack, but maximize the target confidence $P(t | x')$.
4. Stop when the classifier prediction is no longer y .

The lock is deliberately irreversible. Releasing it would reintroduce the exploration overhead that OTS is meant to remove. Across fixed switches $T \in \{5, \dots, 500\}$, performance is statistically indistinguishable, so the useful class ranking stabilizes very early.

3. Experimental Protocol

Models	AlexNet, ResNet-18, VGG-16, ResNet-50, ViT-B/16
Attacks	SimBA, Square Attack with CE loss, Bandits
Modes	Untargeted baseline, targeted oracle, OTS
Budget	$\epsilon = 8/255$ in L_∞ , 10K iterations
Scale	100 ImageNet images per model, 4,500 standard-model runs
Metrics	Success rate and censored mean iterations

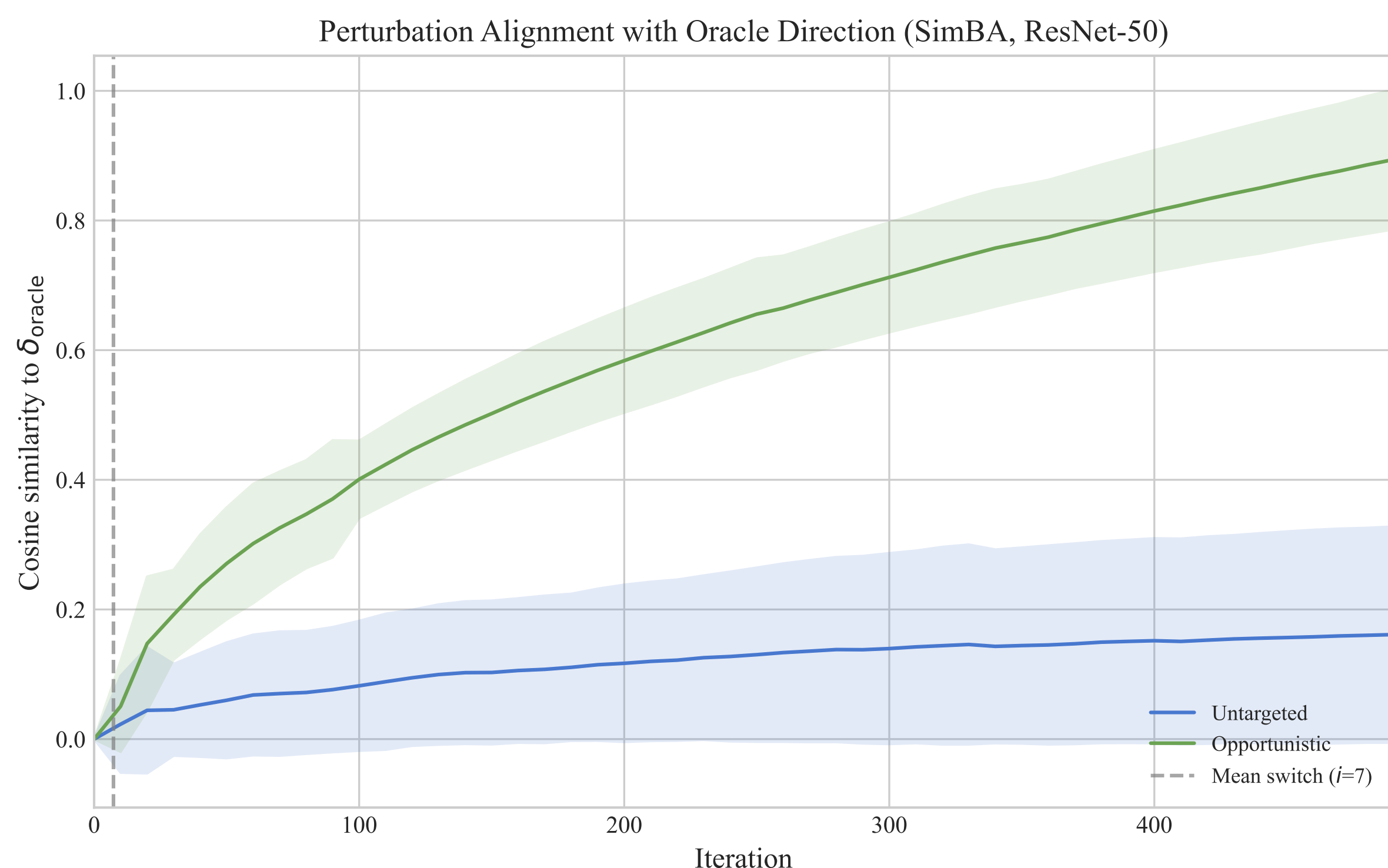
Oracle target. The oracle is the class reached by a same-seed untargeted probe. It is a trajectory-specific ceiling, not a deployable method.

4. When Should OTS Help?

Attack	Direction source	Expected OTS effect
SimBA	No target tracking	Large gain from early commitment
Square CE	CE suppresses y only	Gain similar to a margin surrogate
Bandits	Gradient estimate	Small or negative gain

The negative Bandits case is part of the result, not a failure of tuning: once an attack already estimates a useful direction, imposing a fixed target can interfere with that direction.

5. Mechanism



After lock-in, OTS aligns perturbations with the oracle direction. On SimBA/ResNet-50, terminal cosine similarity rises from 0.174 for untargeted perturbations to 0.865 for OTS.

Interpretation. Once OTS fixes t , maximizing $P(t | x')$ approximates the competitor term of margin loss:

$$\mathcal{L}_{\text{margin}} = f_y(x') - \max_{k \neq y} f_k(x').$$

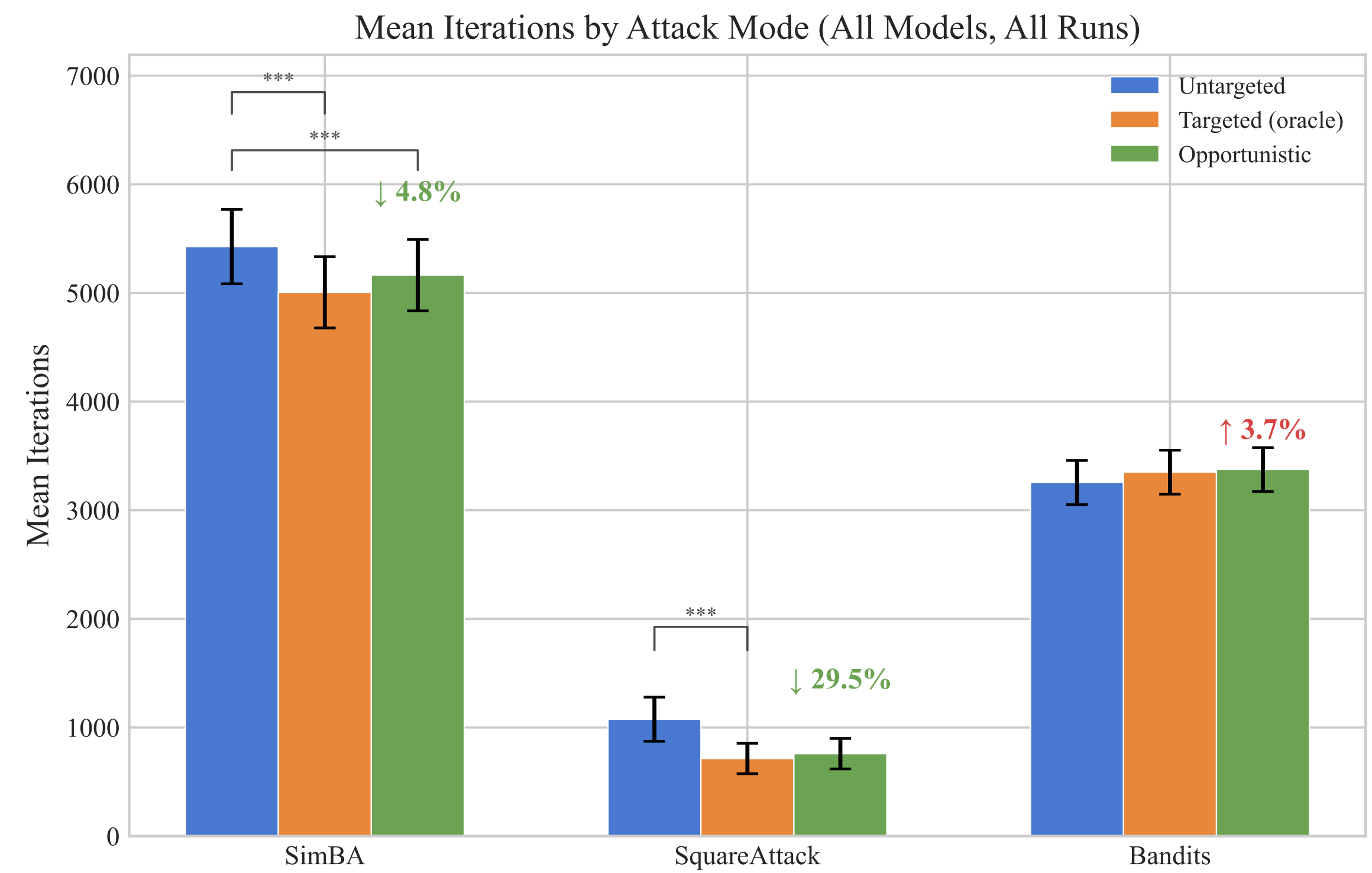
OTS is therefore a lightweight margin-loss surrogate for attacks that do not already track a competitor.

6. Target-Rule Ablation

Target rule	SimBA	Square CE
Random non-true class	14%	53%
Clean-image argmax	79%	96%
Short exploration	80%	99%

Random targets are catastrophic. Clean-image rankings are already strong for SimBA, while Square Attack benefits from a short exploration phase because random patches disturb the initial ranking.

7. Main Results



Attack	Untargeted	OTS	Oracle
SimBA	67%	75%	76%
Square CE	96%	99%	99%
Bandits	37%	34%	35%

+27 pp	43%	0
SimBA success on ResNet-50	relative iteration reduction	gradient access

Main finding. OTS nearly matches the trajectory oracle for SimBA and Square Attack with CE loss, while Bandits gets worse because it already follows an estimated gradient direction.

8. Lock-In Dynamics



In this ResNet-50 example, the faded untargeted trajectory suppresses the true class but then drifts. The vivid OTS trajectory locks onto class 988 and continues to increase that class until the attack succeeds.

Practical point. OTS does not need the exact oracle class in every run; it only needs a viable class that remains competitive after lock-in.

9. Loss Interpretation

Square Attack, ResNet-50	Success	Mean iter.
CE untargeted	85.0%	2,601
CE + OTS	98.0%	1,791
CE oracle	99.0%	1,865
Margin untargeted	98.7%	1,453
Margin + OTS	98.7%	1,583

OTS repairs CE-style drift, but default margin loss already performs dynamic competitor tracking at every iteration. This explains why OTS is useful for CE variants and redundant for margin-based Square Attack.

10. Limits & Future Work

On adversarially trained ImageNet models, success rates are flat across untargeted, oracle, and OTS. The issue is not bad target selection; robust models remove the medium-difficulty regime where directional commitment pays off.

Future work. Test smaller-label regimes such as CIFAR-10, L_2 budgets, randomized defenses, and adaptive re-locking when the selected class drops in rank.

11. Takeaways & Links

Take-away. OTS is a small loss-schedule change: explore briefly, lock onto the strongest non-true class, then optimize that class. It helps drift-prone score attacks and exposes when an attack already has enough directional information.

[1] Guo et al., *Simple Black-box Adversarial Attacks*, ICML 2019.

[2] Andriushchenko et al., *Square Attack*, ECCV 2020.

[3] Ilyas et al., *Prior Convictions*, ICLR 2019.